



Autodesk Fusion Kickstarter Guide

by Henry Li



Introduction

Autodesk Fusion is a cloud-based desktop computer-aided-design (CAD) software for designing & rendering 3D models, simulating designs under different environments, setting up designs for computer-aided-manufacture (CAM) with CNC machinery, designing & simulating electronics. The software also enables engineers to collaboratively work together on projects seamlessly.

Fusion is used in many corporations & industries for designing worldwide best-selling products.

Setting up Fusion on your personal PC / Mac

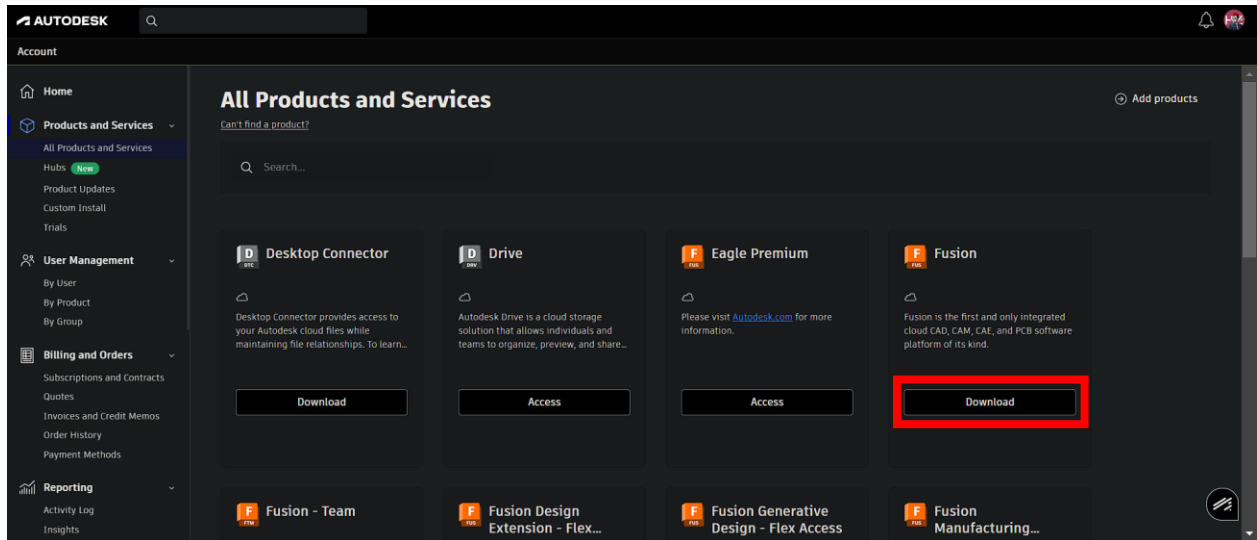
(Please skip this part if you are working in the lab)

1. Go to autodesk.com/education/.
Create an Autodesk account with your RHUL email if do not have one, then login with it. (Please skip to step 5 if you have a verified Autodesk account)
2. Select "Fusion", then the "Student" subscription plan, and "Access products" on the website.
3. Autodesk should now forward you to the student status verification.
Input your full legal name and date of birth.
then for the school information, enter the following details:

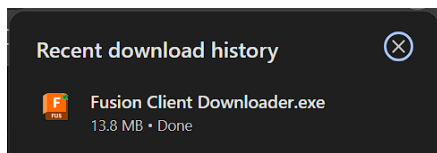
4. You will then be redirected to SheerID, where you will be asked to verify using your university credentials.
Simply verify with your RHUL email address and password and you are all set!



- Once you are verified, you can go back to the initial page, and select "Fusion" again, Autodesk will now direct you to their 'All Products and Services' page, where you can scroll through the list and look for 'Fusion', and click "Download".



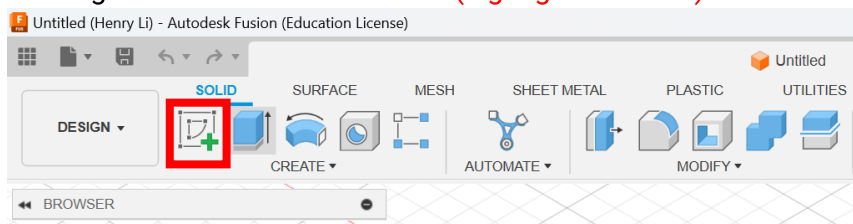
- Run the 'Fusion Client Downloader.exe' and follow the instructions.



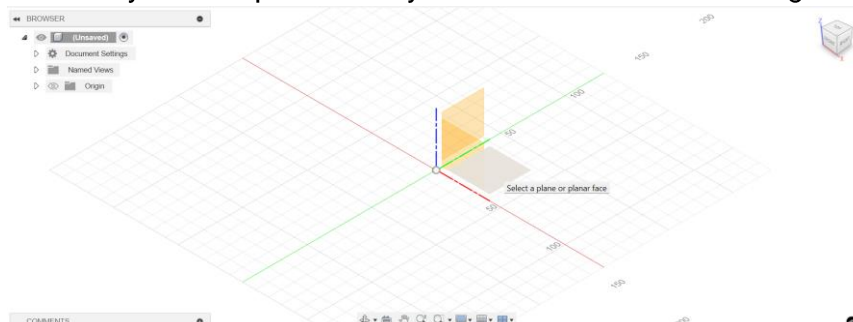
- Once downloaded, Fusion may ask you to login again. So just login again with your Autodesk account details.

Getting started with 3D Modelling

- Fusion will start up with an 'Untitled' CAD design on default settings, so you can start right away by clicking the "Start 2D Sketch" icon (highlighted below)

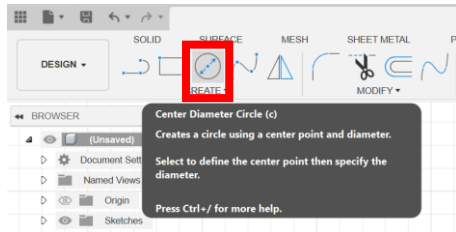


- Select any of the 3 planes that you would like to start working on.

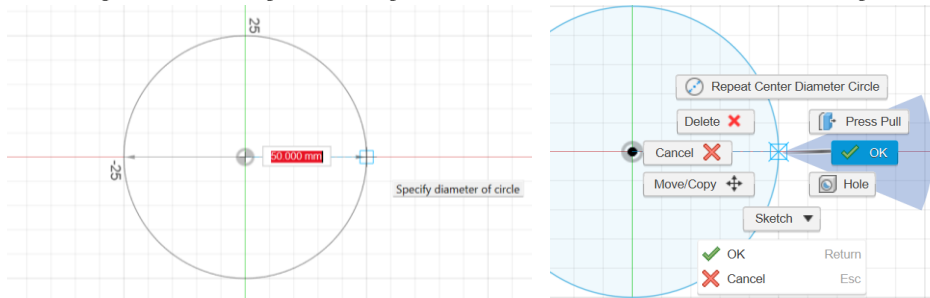




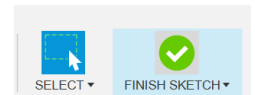
3. Firstly, we will start off with drawing a circle, use the circle tool from the toolbar at the top.



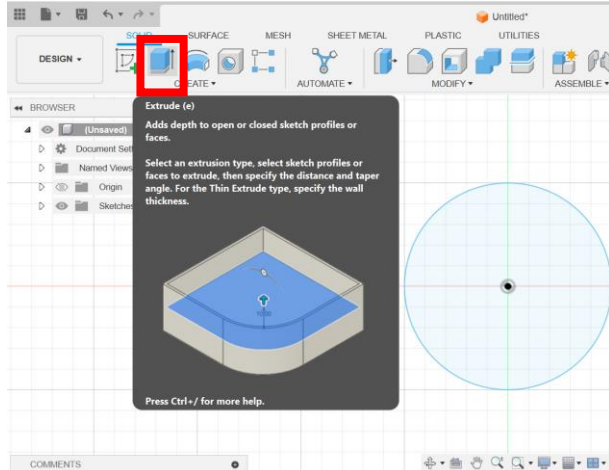
4. Draw a 50.00mm circle from the origin, and left-click on the mouse to confirm it. Then right-click, drag to the right and click "✓ OK" to finish drawing with the shape tool.



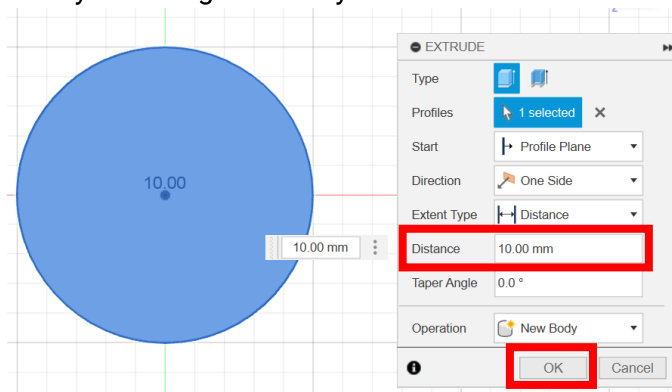
5. Click "Finish Sketch" on the top-right corner on the toolbar to exit the sketching mode.



6. Now using the "Extrude" tool, positioned next to the "Start 2D Sketch" icon, we can extrude the sketched shape that we've just drawn to a solid object.

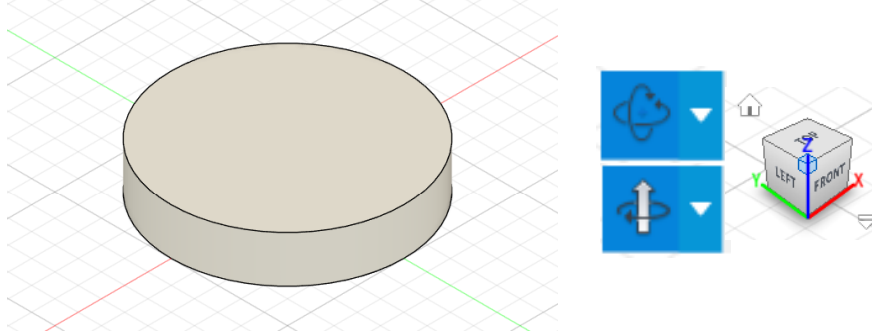


7. Select the circle we just drawn and change the extruding distance to 10.00mm, and click 'OK'. Now you have got a flat cylinder!

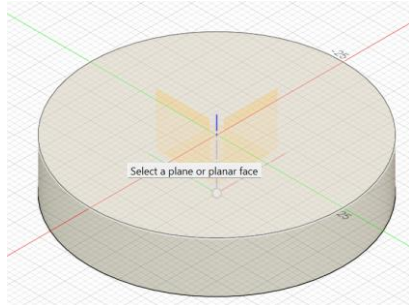




8. You can now use the “Free Orbit” & “Constrained Orbit” on the bottom navigation bar, or the “Orthographic Viewing Cube” at the top-right corner, to adjust your viewing angles to 3D.

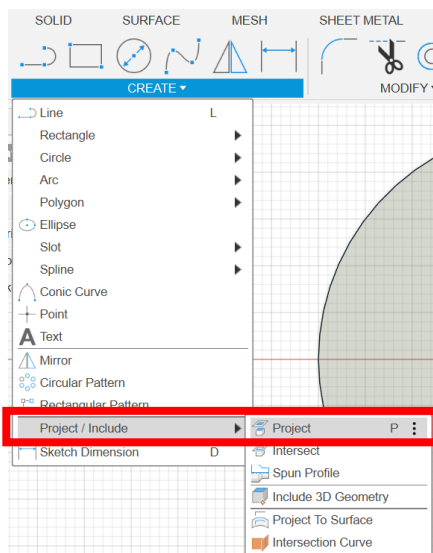


9. Now try and create another 2D sketch on the cylinder's top plane.



10. Select the “Project” tool by going to the drop-down menu “Create” >> “Project / Include” >> “Project (P)”.

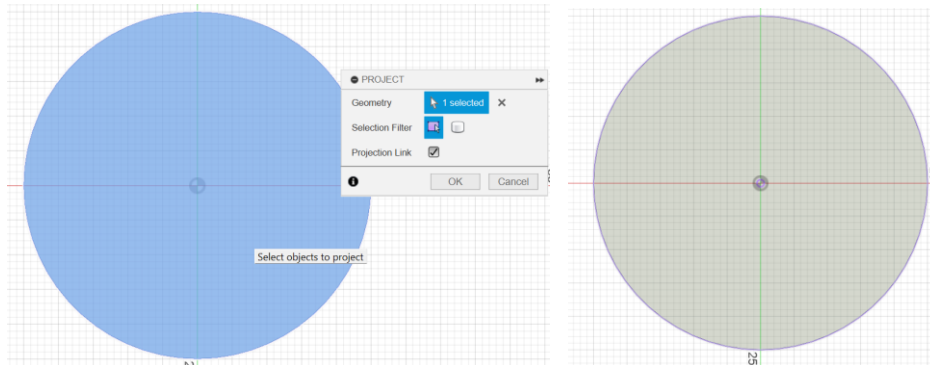
This tool allows you to project and copy any lines or geometries that were previously extruded or drawn in another sketch.



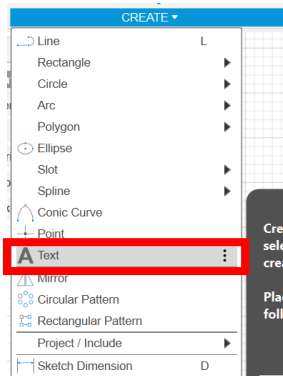


11. Select the circle of the solid cylinder, and click 'OK'.

You should see a purple line on the outline of the circle, which means this line is being projected. And you can start sketching other new lines and shapes according to this projected geometry in this sketch.

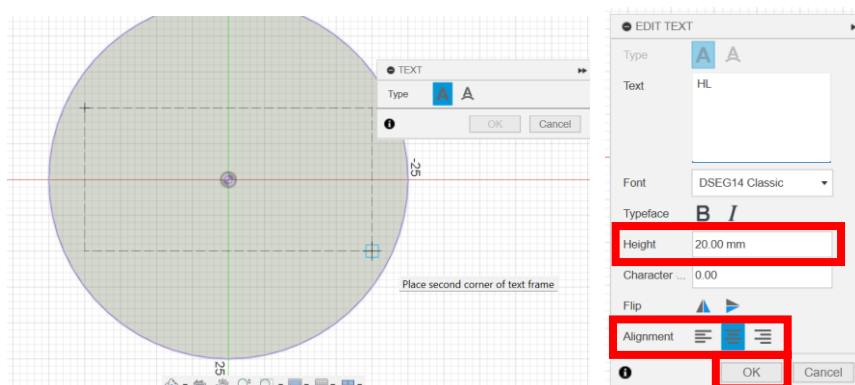


12. Next go to the drop-down menu under "Create" again and select "Text".

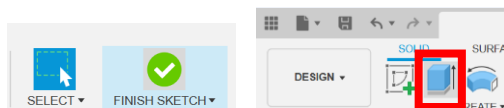


13. Drag the text box across the circle, so that it will be neatly aligned on the circle. Left-click again at the end point after finish dragging the text box to start editing.

Use the properties box on the right to change the font, height, character spacings, and alignments. Enter your initials for the text and set the text height to 20.00mm, and to centre-aligned, as shown in the screenshot below.



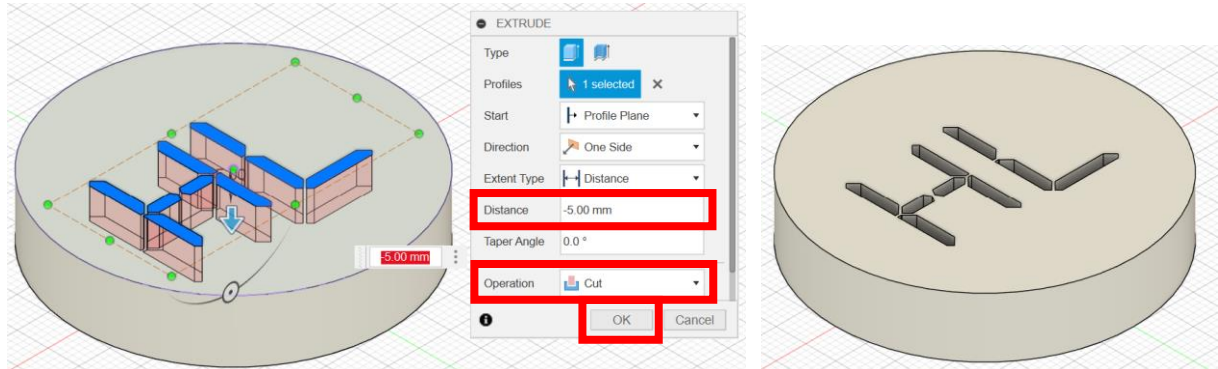
14. Now click "Finish Sketch" and select the "Extrude" tool again.





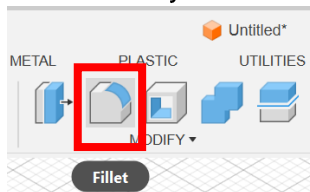
15. This time select the text for extruding, and change the distance to a negative, and select 'Cut' as the operation.

This will make the text as a cut out from the cylinder.



16. We will then make some slight improvements on the overall shape to make it more aesthetically pleasing.

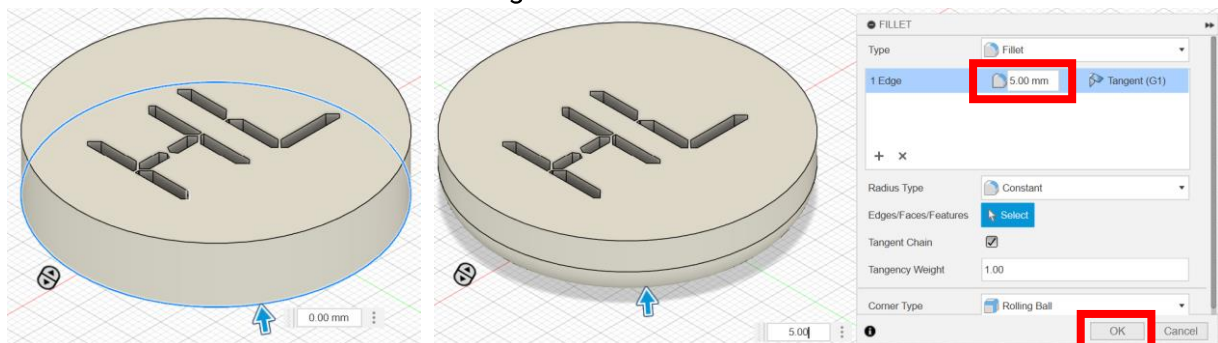
We will first be using the "Fillet" tool, which is located at the centre of the toolbar, in the 'Modify' section.



17. Select the base edge of the cylinder for filleting.

Change the filleting radius to 5.00mm, and click 'OK'.

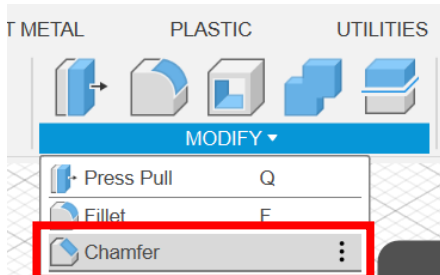
This allows us to rounds some bold edges or corners to some round and smooth curves.



18. Lastly, we will be using another tool from the 'Modify' section on the toolbar.

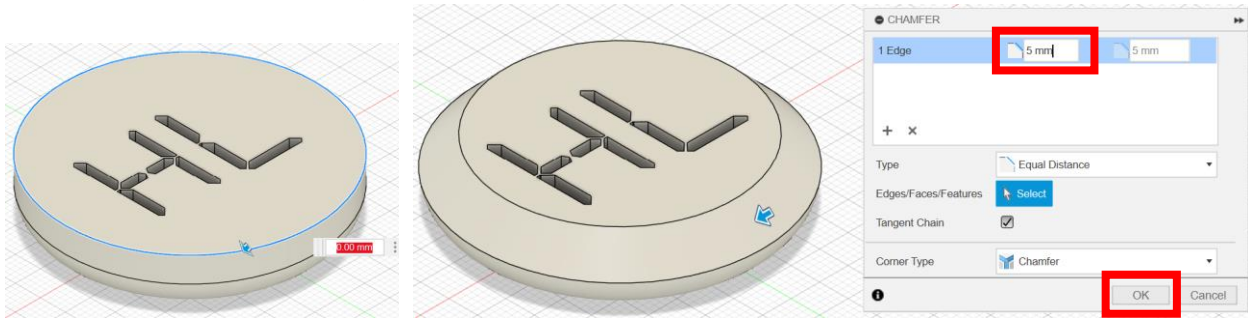
Just open up the drop-down menu, and select the "Chamfer" tool.

Unlike the "Fillet" tool, the "Chamfer" tool bevels these edges to create a diagonal sliced edge.





19. Select the top edge of the cylinder, and also set the chamfering distance to 5.00mm, and click 'OK'.



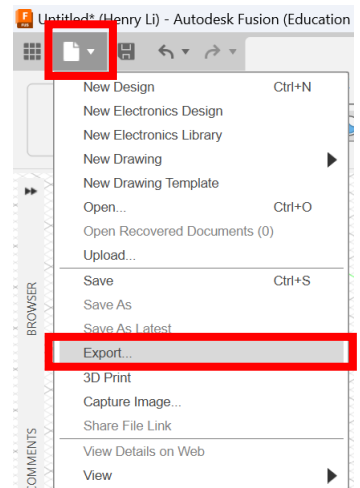
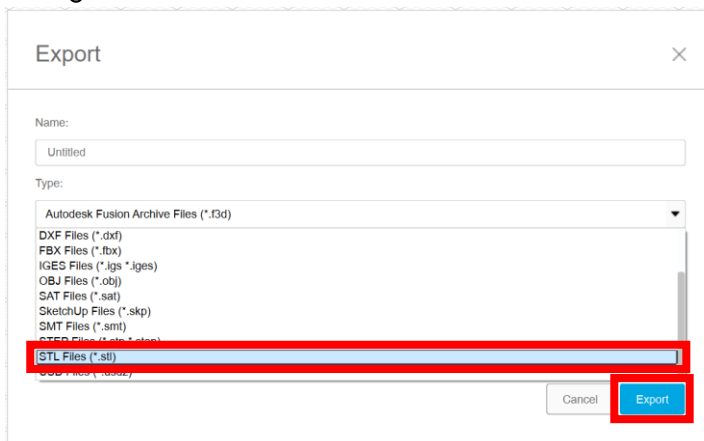
Extended Task – Try adding a key hole at the top-centre position of your token, so that you can use it as a keyring.

Be Creative! – Try and make something else with other geometric tools like the “Rectangle” or even the “Polygon” tools too!

20. Now you have a completed token with your initials on it, let's export it and get it prepared for 3D printing!

Exporting the design to an STL file

1. Go to 'File', at the top-left corner of the window, select “Export”.
2. On the pop-up window, change the file type to 'STL Files (*.stl)', and change the filename to “YourName_Date_MakersToken”.



3. Finally, click “Export”, and send the STL file to makers@royalholloway.su, with file name in the title, and the content: “3D printing, Token, Colour”.
(Please note that you may not have an option for colours depending on the day, please ask a member of the committees)